

Workspace Constraints in Classical Cueing

Craig Brown¹

(1) Director, BrownSim Ltd, London, TW8 9DN, e-mail: craig.brown@brownsim.io

Abstract - Classical cueing, and other “static” filters, benefit from simplicity and popularity in industry. However, workspace constraints are not incorporated in these approaches—there is no mechanism preventing workspace constraint violations.

On the other hand, Model Predictive Control (MPC) is an effective solution to the workspace and cueing problem. Indeed, the cueing problem can be thought of as a constrained double integration, a linear model which fits well into the MPC paradigm.

Disadvantages of MPC are that it is difficult to tune and requires expert knowledge to implement correctly. Maintaining consistent cueing “character” across different scenarios and vehicles is harder compared with the classical approach. In the author’s experience, many elite drivers prioritise “consistency” over “cueing quality”.

It is therefore desirable to augment Classical Washout Filters (CWF’s) with workspace constraint functionality. An approach for doing so is given here.

Keywords: Classical, Cueing, Workspace, Limiting, Limiter

Introduction

Workspace constrained MPC approaches, many of which incorporate kinematics model and/or perception models, are well established. Examples in literature include Asadi, et al. (2023), Bruschetta, Maran, and Beghi (2017), Bruschetta, Cenedese, and Beghi (2019), Bruschetta, et al. (2016; 2018), Fang and Kemeny (2014), Fang, Wautier, and Kemeny (2019; 2021), Garrett and Best (2013), Khusro, et al. (2020), Lamprecht, et al. (2019), Pham, Pham, and Nguyen (2023), and Qazani, et al. (2021).

These approaches are, to varying extents, difficult to tune and require expert knowledge to develop. Indeed, their adoption in industry severely lags their maturation in academia (Ghafarian, et al., 2023).

Classical Washout Filter (CWF’s), on the other hand, are easy to tune, easy to implement (Qazani, et al., 2024) and produce acceptable outcomes. Their simplicity makes them suited for “dynamic” simulator sessions where driver preferences, scenarios and car setup are changing rapidly. Keeping the simulator within workspace constraints is a challenge however.

A known method for workspace limiting in CWF Motion Cueing Algorithms (CWF-MCA’s) is clipping actuator motions. Consider the case of naïvely clipping position, velocity and acceleration of an actuator. Hard clipping position introduces violations (discontinuities) in velocity and acceleration. If these are clipped subsequently, the position constraint may no longer be satisfied. An advantage of hard clipping however is the clear division between “normal driving” and a “limiting event”. In the author’s experience, many elite drivers prefer this over a more ambiguous combination of cueing and limiting (e.g. MPC).

Nonetheless, to soften discontinuities, soft clipping curves needs to be designed. This greatly complicates the parametrisation of a motion simulator. Even if implemented correctly, the effect of soft clipping a single actuator on the overall motion may introduce additional accelerations not accounted for by the vehicle model, for example, an artificial rotation caused by a single actuator reaching its limits. Moreover, the softer the clipping, the more it degrades the motion in general. Finally, any clipping may adversely interfere with recentering (washout). Additional recovery logic may therefore need to be implemented.

Clipping in world space (rather than actuator space) is also an option. But this is complicated by the trend for redundant degrees of freedom, for example, the popular 6 Degree of Freedom (DOF) + XY(ψ) configuration (Ghafarian, et al., 2023).

Nowadays, several simulators of different physical design may coexist in the same simulator facility (Kolff, et al., 2022). Consider MPC—understanding the kinematics of each system, the possible process of its mathematical simplification, linearisation and design of a cost function is a heavy burden for commercial simulator teams. Greater productivity and transferability may be achieved if each simulator can be treated in the same way with respect to limiting.

Clearly, a new philosophy to workspace limiting is needed that is simple enough for day to day operation and provides a unified workflow for a large simulator centre.

Separating Limiting from Kinematics

Kinematics vary from simulator to simulator. For a unified workspace limiter it is critical, then, that limiting occurs on e.g. driver position in an inertial (world) coordinate system—convenient since washout filters typically also act in this reference frame (Brown, 2024b). This way the same limiting algorithm can be used for each simulator, albeit with different mappings (explained subsequently).

If the shape of the workspace in world coordinates is unknown then a mapping f is needed from simulator states to workspace limits. For example, if the simulator is stationary and in the centred position, the driver/cockpit's acceleration limits in each of the 6 degrees of freedom may be high. If the simulator is in an offset position, acceleration limits may be reduced in that direction. This likely reflects zoning in an underlying safety-critical motion control system (Scalera, et al., 2020).

Let $\mathbf{X}_a$ be simulator actuator states and $\mathbf{Y}_+ \in \mathbb{R}^n$, $\mathbf{Y}_- \in \mathbb{R}^n$ be the maximum and minimum allowed values for n driver/cockpit states (position, velocity, acceleration, etc...) in all 6 degrees of freedom (surge, sway, heave, roll, pitch and yaw). Especially for overdetermined systems it is desirable to also

consider the current driver/cockpit state $\mathbf{Y}$. This information is useful for managing antagonistic demands for actuator motion from conflicting degrees of freedom (roll and heave, for example).

In summary it is desirable to *design*:

$$f : \mathbf{X} = \begin{pmatrix} \mathbf{X}_a \\ \mathbf{Y} \end{pmatrix} \rightarrow \begin{pmatrix} \mathbf{Y}_+ \\ \mathbf{Y}_- \end{pmatrix} = \mathbf{Y}_{\text{lim}} \quad (1)$$

The level of detail implemented in this mapping may vary depending on resource and budget constraints—either, capturing every nuance or, providing an approximate shape to the workspace—perhaps even based simply on catalogue values.

The implementation of the mapping may be mathematical, based on a kinematics model. Alternatively, it may be based on an empirically populated Look up Table (LUT)—interpolation should be used as much as possible to minimise memory usage. LUT's or Piecewise Affine Functions (PWA's) are a known approach to improving computational efficiency of control algorithms, especially MPC (Chadha, et al., 2023; Fang and Kemeny, 2012; Munir, et al., 2017)

Indeed, Bruschetta, et al. (2014) concluded that determining the analytical shape of a 6+3 DOF simulator workspace is too difficult in practice. A curve fitting procedure was used on a set of measurements instead. According to Bessala, Bidaud, and Ben Oueddou (1996) “explicit direct or inverse models [of Stewart platform workspaces]... are not in general analytically possible.” Some visualisations of workspaces in world coordinates are given in Grottole, et al. (2018).

The exact method of construction of the LUT is beyond the scope of this paper—is at the discretion of the practitioner. Interpolation and simplifications of the workspace should likely be used to manage its size in memory.

The LUT may be used to dynamically parametrise some bound or linear constraints. An analogous approach is given in Song and Lau (2022) where a workspace in actuator space is “convexified” and passed to an MPC.

In any case, an approximate process for creating a mapping is given below. Let $\mathbf{X}_a^*$ be the set of all sampled points in the actuator space (which should be as small as possible) for a given simulator and let $\mathbf{Y}^*$ be the same for driver states. All non-sampled states are mapped using interpolation or extrapolation.

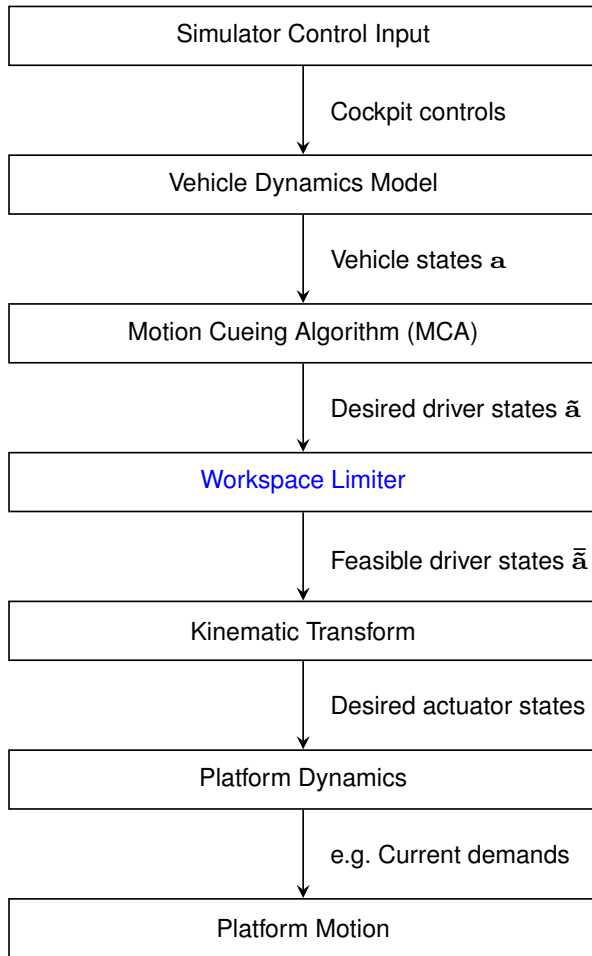


Figure 1. Modified simulator control flow chart from Telban and Cardullo (2005). The proposed workspace limiter is a separate subsystem that acts on head or h-point position, velocity, acceleration etc... $\in \mathbf{Y}$. This strict separation facilitates more experimentation in MCA design.

Algorithm 1: Populate workspace mapping LUT

```

1 for  $s \in \text{simulators}$  do
2   for  $\mathbf{X}_a \in \mathbf{X}_a^*, \mathbf{Y} \in \mathbf{Y}^*$  do
3      $\text{LUT}[s][“x”].\text{append}(\begin{pmatrix} \mathbf{X}_a \\ \mathbf{Y} \end{pmatrix})$ 
4     if kinematics model available then
5       Determine  $\mathbf{Y}_{\text{lim}}$  using kinematics model
6     else
7       Experimentally determine  $\mathbf{Y}_{\text{lim}}$  by perturbing simulator about  $\mathbf{X}_a$ 
8      $\text{LUT}[s][“y”].\text{append}(\mathbf{Y}_{\text{lim}})$ 
9   end
10 end
  
```

Workspace Limiter

Assuming a method of retrieving workspace limits of the driver moving in world space is available, a limiting method that fits within the signal flow of Figure 1 is established. Especially the classical cueing case.

It is desirable to retain the advantages of classical cueing (described in the Introduction) whilst introducing the workspace constraint handling functionality of MPC.

To that end, a simple version of MPC is appended to the classical motion cueing filter, or other MCA that requires workspace limiting functionality. Here, the MPC is used as a filter i.e. the system states are known exactly.

Moreover, the MPC is only used for its constraint handling functionality. This means there is significantly reduced tuning burden compared to a full MPC-MCA.

In the proposed approach, the output of the classical cueing filter is used as the reference or target signal. As long as the simulator is operating within workspace constraints, the classical filter output is unmodified, that is to say, $\tilde{a} = \tilde{a}$. In this regard the approach is similar to Jain, et al. (2023) but differs in that we avoid Nonlinear MPC (NMPC), do not rely on the NMPC for washout (instead closing the loop with a PD controller) and do not include any specific kinematics in the MPC. Another difference here is that the MPC is used merely as a filter—platform control is considered a separate subsystem in line with the signal flow proposed in Telban and Cardullo (2005).

In the event constraints are or will be violated, the MPC will modify the output acceleration accordingly. This causes velocity and position to also deviate from that of the classical cueing filter. Therefore, it is necessary to restore the system again to a state where the classical cueing filter output is reproduced without modification. This is achieved by closing the loop with a Proportional Derivative (PD) controller, as shown in Figure 2. The stability properties for the PD control of double integrator systems are well understood (Hu and Zhang, 2014; Rao and Bernstein, 2001; Tyan and Bernstein, 1999). Moreover, the above is equivalent to a washout filter anyway (Brown, 2023).

MPC

The linear discrete time approximation of the double integrating system is given in Equations 2, 3 and 4.

$$\begin{pmatrix} x(t+h) \\ v(t+h) \\ a(t+h) \end{pmatrix} = \mathbf{A} \begin{pmatrix} x(t) \\ v(t) \\ a(t) \end{pmatrix} + \mathbf{B} \mathbf{u} \quad (2)$$

$$\begin{pmatrix} \tilde{x}(t) \\ \tilde{v}(t) \\ \tilde{a}(t) \end{pmatrix} = \mathbf{C} \begin{pmatrix} x(t) \\ v(t) \\ a(t) \end{pmatrix} \quad (3)$$

$$\mathbf{A} = \begin{pmatrix} 1 & h & 0.5h^2 \\ 0 & 1 & h \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{C} = \mathbf{I} \quad (4)$$

For the sake of simplicity manipulated variable move

suppression (jerk penalty) is not used in this study. A Quadratic Programming (QP) problem, the MPC solves:

$$\begin{aligned} J_y(z_k) = & \sum_{j=1}^{n_y} \sum_{i=1}^p \left\{ \frac{w_{i,j}^y}{s_j^y} [r_j(k+i|k) - y_j(k+i|k)] \right\}^2 \\ & + \sum_{j=1}^{n_u} \sum_{i=0}^{p-1} \left\{ \frac{w_{i,j}^u}{s_j^u} [u_j(k+i|k) - u_{j,t}(k+i|k)] \right\}^2 \\ & + \rho_\epsilon \epsilon_k^2 \end{aligned} \quad (5)$$

$$z_k = [u(k|k)^T \ u(k+1|k)^T \ \dots \ u(k+p-1|k)^T \ \epsilon_k]^T \quad (6)$$

where k is the current control interval, p is the prediction horizon, n_y is the number of plant output variables, $y_j(k+i|k)$ is the i th predicted value of output variable j , $r_j(k+i|k)$ the equivalent reference value, s_j^y scale factor, $w_{i,j}^y$ tuning weight, ϵ_k slack variable, n_u is the number of manipulated variables, $u_j(k+i|k)$ is the i th calculated value of manipulated variable, $u_{j,t}(k+i|k)$ its target value (here 0), s_j^u manipulated variable scale factor, $w_{i,j}^u$ manipulated variable tuning weight and ρ_ϵ penalises soft constraint violation ϵ_k .

For the application of MPC as a limiter, it is important that ρ_ϵ has a high magnitude relative to other weights. Indeed, though in principle the QP solver may consider hard constraints, in practice this may not always be achievable (especially in an interactive setting where the future is difficult to predict). Of the other weights, acceleration error should be penalised the most—recall the goal is to pass through the input MCA unchanged most of the time.

This makes the limiter easy to set up. The weight for acceleration is set an order of magnitude higher than that of velocity and position. The penalty term ρ_ϵ is set many orders of magnitude higher. P and D are set to very conservative values (producing corrective motion that is insignificant in terms of driver perception).

Put another way, the order of priority is first, meeting workspace constraints, second, matching acceleration of the CWF and thirdly, also matching velocity and position (if possible).

The practitioner then only has to consider the prediction horizon, with higher values implying a "soft clipping" approach and lower values a "hard clipping" approach.

Experiments

A driving maneuver is recorded from BeamNG (Figure 3). More information about BeamNG vehicle dynamics simulation is given in Papamichail, Stark, and Gambi (2025). For simplicity, uniform workspace limits are taken from catalogue values of a recent simulator—1.3m, 1.5m s^{-1} and 15m s^{-2} for displacement, velocity and acceleration respectively (DMG-1 2025). The scenario is initially scaled at 60%, the baseline case.

An MPC-MCA and a Workspace Limited Classical

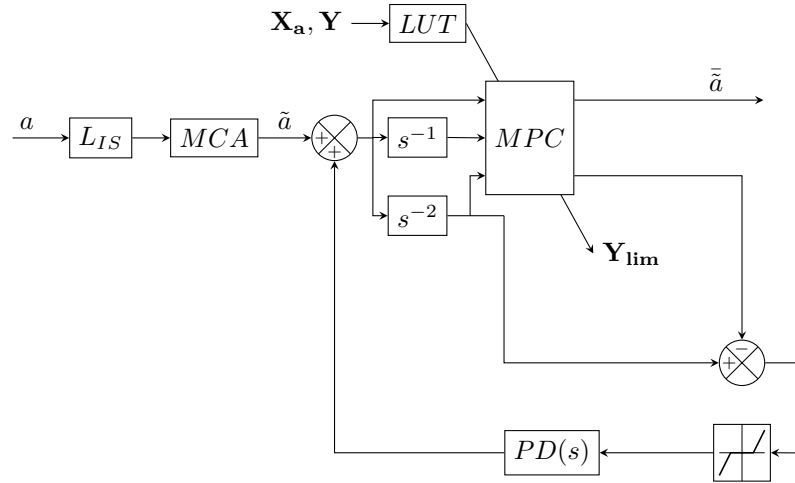


Figure 2. Proposed CWF with workspace limiter. So long as the CWF is reasonably well tuned, the limiter will pass through the cueing demands unchanged most of the time. The MPC receives a reference signal of acceleration, velocity and position from the CWF. In the case workspace constraints prevent these being met, the feedback system consisting of a PID will slowly restore simulator alignment to the CWF requests. The deadzone optionally may be used to reduce the effect on overall motion, at the expense of simulator offset.

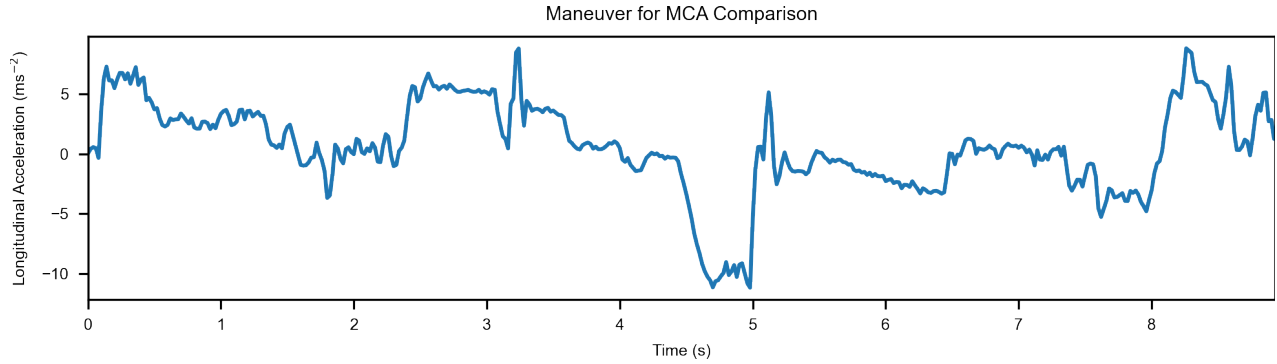


Figure 3. Acceleration time series for study.

Washout Filter (WLCWF) are prepared for comparison. The MPC is of the simplest type applicable to motion cueing, with no kinematics or perception model. However, the QP solver is made aware of the aforementioned uniform workspace constraints.

For the MPC, following the recommendation of van der Ploeg, et al. (2020), the input error weight is set to a small value $w^u = 0.1$. Additionally, the reference value is constant throughout the control horizon, matching its value at the current time. For velocity and position this is set to 0 (encouraging washout). A constant reference is more representative of “on-line” driver in the loop simulator use, especially where the driver is “unpredictable”. The same is true for the MPC of the WLCWF, only that the velocity and displacement references come from the CWF instead.

The remaining parameters, including control horizon (Katliar, et al., 2015), are tuned with a recent derivative free, global search routine based on Brown, et al. (2016) and Zhang and Sanderson (2009) within the ProCue environment (Brown, 2024a). The final parameters are $p = 41$, weights 3.79, 1.26 and 5.94 for acceleration, velocity and displacement respectively.

For the WLCWF, the limiting is initially disabled and the classical washout parameters are fitted to the vehicle data scaled at 60%. This provides a non-limited reference that fully uses simulator travel. The final parameters for the CWF (Equation 7) are $\delta = 1.58$, $\omega_0 = 0.21$ and $\omega_1 = 4.06$. For the limiter, $p = 20$,

weights 10, 1, 1 for acceleration, velocity and displacement error respectively, $w^u = 0.0$, $K_d = 0.5$ and $K_p = 0.1$. The soft constraint penalty $\rho_\epsilon = 1e6$.

$$G(s) = \left(\frac{s^2}{s^2 + 2\delta\omega_0 s + \omega_0^2} \right) \left(\frac{s}{s + \omega_1} \right) \quad (7)$$

For subsequent experiments, the WLCWF and MPC are ‘detuned’ by upscaling vehicle motion to 80% and 100%. This represents the fact that in commercial applications, there is rarely time to optimise cueing parameters between sessions. In other words, it is usual to find MCA parameterisation is sub-optimal in practice. Other ways a cueing algorithm can become detuned is if the driver becomes more aggressive over time (as they build confidence) or if the starting position of the simulator drifts between laps or repetitions of a scenario.

In any case, since limiting functionality is the focus of this study, we focus on the case where motion is “too much” for the workspace. A time step of $0.01s$ is used for both.

Results

Firstly, the known result that well-tuned MPC outperforms a well-tuned CWF is re-established. Though

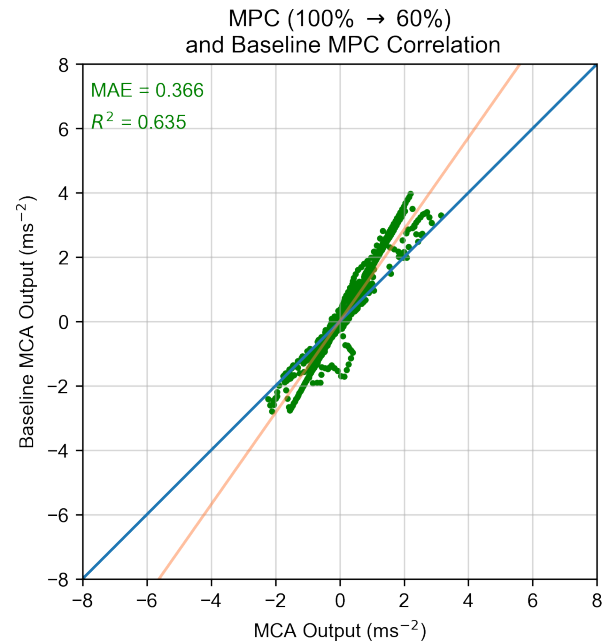
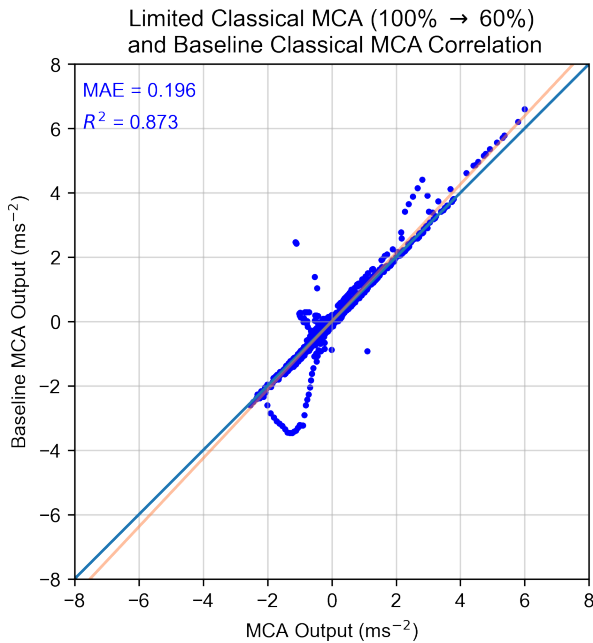
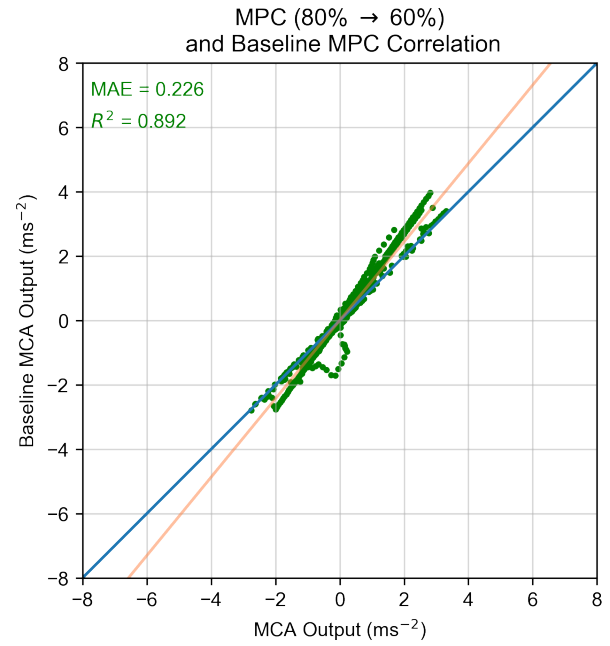
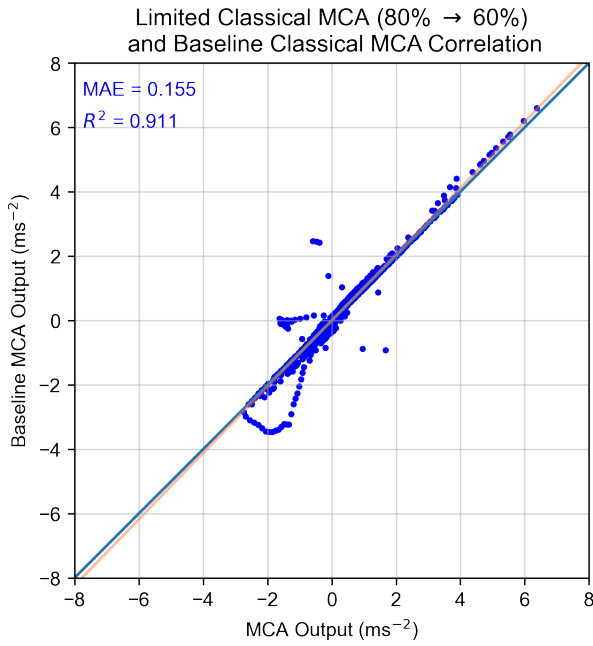


Figure 4. Self-correlation of WLCWF wrt. detuning against reference $y = x$ line.

Figure 5. Self-correlation of MPC wrt. detuning against reference $y = x$ line.

both MCA's reach workspace limits the MPC produces more motion in general, which is consistent with previous work. This translates to better correlation with the simulated vehicle accelerations (Figure 6). To quantify this, Mean Absolute Error (MAE) and coefficient of determination (R^2) is used (Schneider and Xhafa, 2022).

The effect of detuning the MCA's is then investigated for the WLCWF (Figure 4) and MPC (Figure 5). In summary, the WLCWF is better able to retain its best fit line and self-correlation as it is scaled into the limiter. Put another way, its character is somewhat retained even as the limiter is used more.

The MPC is also able to accommodate a detuned system due to its innate handling of constraints. Whereas WLCWF reproduces, in the most part,

scaled motion from the baseline case, the MPC shifts the motion delivered more dramatically. It should be said that putting a higher weight on position error could help the MPC behave more like a CWF, providing slack to accommodate increased motion. This is a known consideration for practitioners carrying out offline optimisation of MPC-MCA parameters.

Indeed, our naïve offline optimisation process tries to minimise motion error. With an arbitrarily complex cost function, we surely could see MPC and CWF behaviour converge. Again, the merits of MPC are well known in the prior art, and it is not the author's intention to diminish these.

In any case, it is clearly possible to achieve MPC-like limiting with CWF-like ease of tuning. And, this limiting is somewhat "stable in character" in the face

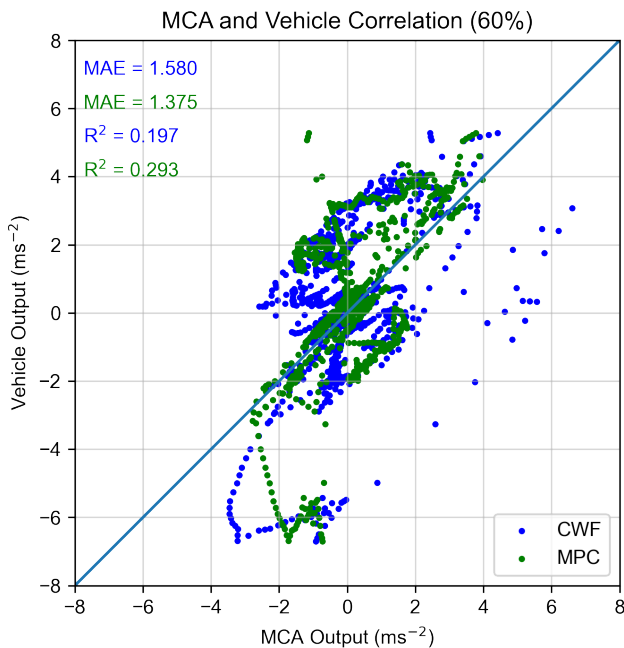


Figure 6. Consistent with existing literature, MPC outperforms CWF.

of severe detuning. The disadvantage of (WL)CWF compared to MPC, less motion overall, remains.

Further Work

This paper presented a proof of concept workspace limiter that may be placed downstream of any MCA (including CWF). Further work is needed to validate its benefits with subjective driver feedback and refine the mapping process given in Algorithm 1.

Further work is also needed to understand the best approach to updating MPC constraints dynamically, especially in systems where workspace constraints, e.g. velocity, change rapidly as a function of position or orientation.

Finally, other states such as jerk and absement are highly relevant to simulator workspaces. The former is limited by the mechanical design of the simulator. The latter reflects constraints with respect to overheating of (now common) direct drive systems in certain orientations. Indeed, perhaps future systems should incorporate a quadruple rather than double integrator, $j \rightarrow A$.

Conclusion

In this paper a new approach to workspace limiting was introduced that is intended as an upgrade for CWF and similar, and, like CWF, is applicable irrespective of simulator kinematics (after a one-off process to establish the shape of the workspace in world coordinates).

Indeed, like CWF's, the limiter works on driver/cockpit location in an inertial reference frame, rather than on actuator positions in an actuator reference frame.

MPC is repurposed as an enhanced clipping method. Unlike conventional clipping, which may introduce discontinuities, constraints are handled elegantly as

in MPC-MCA. However, unlike in MPC-MCA, the output of the CWF is used as the reference for acceleration, velocity and position. An additional PD controller is needed to ensure motion re-converges to the CWF reference after a limiting event.

The effectiveness of the approach is highly dependent on the design of the mapping, especially since actuator positions are unknown to the MPC limiter. However, the proposed signal flow could allow for actuator positions to be incorporated into a (N)MPC limiter, if desired. This removes the need to design a mapping.

References

- Asadi, H., Bellmann, T., Qazani, M. C., Mohamed, S., Lim, C. P., and Nahavandi, S., 2023. A Novel Decoupled Model Predictive Control-Based Motion Cueing Algorithm for Driving Simulators. *IEEE Transactions on Vehicular Technology*, 72(6), pp. 7024–7034.
- Bessala, J., Bidaud, P., and Ben Ouezdou, F., 1996. Analytical study of Stewart platform workspaces. In: *Proceedings of IEEE International Conference on Robotics and Automation*. Vol. 4, pp. 3179–3184.
- Brown, C., 2023. A Nonlinear Extension to Classical Filters for Washout Miscue Prevention. In: A. Kemeny, J.-R. Chardonnet, F. Colombet, S. Espié, and F. Merienne eds. *Proceedings of the Driving Simulation Conference 2023 Europe VR*. Driving Simulation Association. Antibes, France.
- 2024a. ProCue: State-of-the-art Motion Cueing For All. In: A. Kemeny, J.-R. Chardonnet, F. Colombet, S. Espié, and F. Merienne eds. *Proceedings of the Driving Simulation Conference 2024 Europe VR*. Driving Simulation Association. Strasbourg, France.
- 2024b. Reduced Cross-Axis Distortion Motion Cueing. In: A. Kemeny, J.-R. Chardonnet, F. Colombet, S. Espié, and F. Merienne eds. *Proceedings of the Driving Simulation Conference 2024 Europe VR*. Driving Simulation Association. Strasbourg, France.
- Brown, C., Jin, Y., Leach, M., and Hodgson, M., 2016. μ JADE: adaptive differential evolution with a small population. *Soft Computing*, 20, pp. 4111–4120.
- Bruschetta, M., Maran, F., and Beghi, A., 2017. A fast implementation of MPC-based motion cueing algorithms for mid-size road vehicle motion simulators. *Vehicle System Dynamics*, 55(6), pp. 802–826.
- Bruschetta, M., Cenedese, C., and Beghi, A., 2019. A real-time, MPC-based Motion Cueing Algorithm with Look-Ahead and driver characterization. *Transportation Research Part F: Traffic Psychology and Behaviour*, 61. Special TRF issue: Driving simulation, pp. 38–52. ISSN: 1369-8478.
- Bruschetta, M., Maran, F., Beghi, A., and Minen, D., Sept. 4, 2014. An MPC based Motion Cueing Algorithm with side slip dynamics. In: A. Kemeny, F. Colombet, F. Merienne, and S. Espié eds. *Proceedings of the Driving Simulation Conference 2014 Europe*.
- Bruschetta, M., Maran, F., Cenedese, C., Beghi, A., and Minen, D., Sept. 7, 2016. An MPC-Based Motion Cueing implementation with Time-Varying Prediction and drivers skills characterization. In: A. Kemeny, F. Merienne, F. Colombet, and S. Espié eds. *Proceedings of the Driving Simulation Conference 2016 Europe*. Driving Simulation Association. Paris, France, pp. 11–20.
- Bruschetta, M., Mendola, D. L., Beghi, A., and Minen, D., Sept. 5, 2018. An MPC based Motion Cueing Algorithm with side slip dynamics. In: A. Kemeny, F. Colombet, F. Merienne, and S. Espié eds. *Proceedings of the Driving Simulation Conference 2018 Europe VR*. Driving Simulation Association. Antibes, France, pp. 55–59.
- Chadha, A., Jain, V., Lazcano, A., and Shyrokau, B., 2023. Computationally-efficient Motion Cueing Algorithm via Model Predictive Control. English. In: *Proceedings - 2023 IEEE International Conference on Mechatronics, ICM 2023*. United States: IEEE.

- DMG-1 2025. <https://www.dynisma.com/dmg-1>. Accessed: 2025-04-14.
- Fang, Z. and Kemeny, A., 2012. Motion Cueing Algorithms for a Real-Time Automobile Driving Simulator. In: A. Kemeny ed. *Proceedings of the Driving Simulation Conference 2012 Europe*.
- Fang, Z. and Kemeny, A., 2014. Review and Prospects of Renault's MPC Based Motion Cueing Algorithm for Driving Simulator. In: A. Kemeny, F. Merienne, and S. Espié eds. *Proceedings of the Driving Simulation Conference 2014 Europe VR*. Paris, France.
- Fang, Z., Wautier, D., and Kemeny, A., Sept. 4, 2019. Development and applications of a fast MPC based motion cueing algorithm. In: A. Kemeny, F. Colombet, F. Merienne, and S. Espié eds. *Proceedings of the Driving Simulation Conference 2019 Europe VR*. Driving Simulation Association. Strasbourg, France, pp. 109–116.
- Sept. 14, 2021. Renault's new MPC based motion cueing algorithm development. In: A. Kemeny, J.-R. Chardonnet, and F. Colombet eds. *Proceedings of the Driving Simulation Conference 2021 Europe VR*. Driving Simulation Association. Munich, Germany, pp. 149–158.
- Garrett, N. and Best, M., 2013. Model predictive driving simulator motion cueing algorithm with actuator-based constraints. *Vehicle System Dynamics*, 51(8), pp. 1151–1172. <https://doi.org/10.1080/00423114.2013.783219>.
- Ghafarian, M., Watson, M., Mohajer, N., Nahavandi, D., Kebria, P. M., and Mohamed, S., 2023. A Review of Dynamic Vehicular Motion Simulators: Systems and Algorithms. *IEEE Access*, 11, pp. 36331–36348.
- Grottoli, M., Cleij, D., Pretto, P., Lemmens, Y., Happee, R., and Bülthoff, H., Dec. 2018. Objective evaluation of prediction strategies for optimization-based motion cueing. *SIMULATION*, 95, p. 003754971881597. <https://doi.org/10.1177/0037549718815972>.
- Hu, J. and Zhang, H., 2014. Globally asymptotically stable saturated PID controllers for a double integrator with constant disturbance. *International Journal of Robust and Nonlinear Control*, 24(10), pp. 1488–1504.
- Jain, V., Lazcano, A., Happee, R., and Shyrokau, B., 2023. *Motion Cueing Algorithm for Effective Motion Perception: A frequency-splitting MPC Approach*. arXiv: 2309.01689 [cs.RD]. Available at: <https://arxiv.org/abs/2309.01689>.
- Katliar, M., Winkel, K. de, Venrooij, J., Pretto, P., and Bülthoff, H., Sept. 2015. Impact of MPC Prediction Horizon on Motion Cueing Fidelity. In: H. Bülthoff, A. Kemeny, and P. Pretto eds. *Proceedings of the Driving Simulation Conference 2015 Europe*. Tübingen, Germany.
- Khusro, Y. R., Zheng, Y., Grottoli, M., and Shyrokau, B., 2020. MPC-Based Motion-Cueing Algorithm for a 6-DOF Driving Simulator with Actuator Constraints. *Vehicles*, 2(4), pp. 625–647.
- Kolff, M., Venrooij, J., Schwienbacher, M., Pool, D., and Mulder, M., Oct. 2022. Motion Cueing Quality Comparison of Driving Simulators using Oracle Motion Cueing. In: A. Kemeny, J.-R. Chardonnet, F. Colombet, F. Merienne, and S. Espié eds. *Proceedings of the Driving Simulation Conference 2022 Europe VR*. Strasbourg, France.
- Lamprecht, A., Steffen, D., Haecker, J., and Graichen, K., Sept. 4, 2019. Comparison between a Filter and an MPC based MCA in an Offline Simulator Study. In: A. Kemeny, F. Colombet, F. Merienne, and S. Espié eds. *Proceedings of the Driving Simulation Conference 2019 Europe VR*. Driving Simulation Association. Strasbourg, France, pp. 101–107.
- Munir, S., Hovd, M., Fang, Z., Olaru, S., and Kemeny, A., 2017. Complexity reduction in motion cueing algorithm for the ULTIMATE driving simulator. *IFAC-PapersOnLine*, 50(1). 20th IFAC World Congress, pp. 10729–10734.
- Papamichail, C., Stark, D., and Gambi, A., 2025. Scenario-Based Testing with BeamNG.tech (Hands-On Training). In: *2025 IEEE Conference on Software Testing, Verification and Validation (ICST)*, pp. 767–768.
- Pham, D.-A., Pham, T. N., and Nguyen, D.-T., 2023. Novel model predictive control-based motion cueing algorithm for compensating centrifugal acceleration in KUKA robocoaster-based driving simulators. *Science Progress*, 106(4), p. 00368504231204759.
- Qazani, M. R. C., Asadi, H., Mohamed, S., Alqumsan, A. A., Rahimzadeh, G., Lim, C. P., and Nahavandi, S., 2021. An MPC-based Motion Cueing Algorithm Using Washout Speed and Grey Wolf Optimizer. In: IEEE Press.
- Qazani, M. R. C., Asadi, H., Najdovski, Z., Alsanwy, S., Zakarya, M., Alam, F., Ouakad, H. M., Lim, C. P., and Nahavandi, S., 2024. High-fidelity learning-based motion cueing algorithm by bypassing worst-case scenario-based tuning technique. *Cognitive Robotics*, 4, pp. 116–127. ISSN: 2667-2413.
- Rao, V. and Bernstein, D., 2001. Naive control of the double integrator. *IEEE Control Systems Magazine*, 21(5), pp. 86–97.
- Scalera, L., Giusti, A., Cosmo, V., Riedl, M., Vidoni, R., and Matt, D., June 2020. Application of Dynamically Scaled Safety Zones Based on the ISO/TS 15066:2016 for Collaborative Robotics. *International Journal of Mechanics and Control*, 21, pp. 41–50.
- Schneider, P. and Xhafa, F., 2022. Chapter 3 - Anomaly detection: Concepts and methods. In: P. Schneider and F. Xhafa eds. *Anomaly Detection and Complex Event Processing over IoT Data Streams*. Academic Press, pp. 49–66.
- Song, C. and Lau, D., Aug. 2022. Workspace-Based Model Predictive Control for Cable-Driven Robots. *IEEE Transactions on Robotics*, 38, pp. 2577–2596.
- Telban, R. and Cardullo, F., 2005. *Motion Cueing Algorithm Development: Human-Centered Linear and Nonlinear Approaches*. Tech. rep. CR-2005-213747. NASA.
- Tyan, F. and Bernstein, D. S., 1999. Global stabilization of systems containing a double integrator using a saturated linear controller. *International Journal of Robust and Nonlinear Control*, 9(15), pp. 1143–1156.
- van der Ploeg, J., Cleij, D., Pool, D., Mulder, M., and Bülthoff, H., 2020. Sensitivity Analysis of an MPC-based Motion Cueing Algorithm for a Curve Driving Scenario. In: A. Kemeny, J.-R. Chardonnet, and F. Colombet eds. *Proceedings of the Driving Simulation Conference 2020 Europe*. Antibes, France.
- Zhang, J. and Sanderson, A. C., 2009. JADE: Adaptive Differential Evolution With Optional External Archive. *IEEE Transactions on Evolutionary Computation*, 13(5), pp. 945–958.